

Phase 5 Completion Report

Backend Completion, Business Logic Hardening & Full System Verification

Date: August 26, 2026

Scope: Audit-first verification of the entire backend against the LIVE production Supabase database (booking lifecycle, payment lifecycle, availability/double-booking, pricing integrity, security/RLS, Edge Functions, migration drift) — no rebuilding of completed functionality, no work pulled forward from later phases.

Method: Full repo/migration/Edge Function/test audit, plus 12 live functional tests executed directly against production inside rolled-back transactions (zero permanent writes), plus direct read-only inspection of the one real production booking.

Repository: dubai-airport-rental — one new migration file, authored and now applied to production (see §12)

✓ 150/150 tests passing · Build clean · Lint unchanged · Live schema/RLS/grants/function body match migrations exactly · Zero test-data residue in production

✓ Booking status guard migration APPLIED and LIVE-TESTED — 11/11 transition tests pass, including exact reproductions of the real production bug, now correctly blocked. Phase 5 is CLOSED.

1 Executive summary

This phase did not add features — it verified that everything built in Phases 0–4.2 actually works against the real, live production database, and hardened one business-logic gap that verification uncovered. The audit read every migration, Edge Function, shared library, admin API file, and test file in the repo, then tested the booking lifecycle, payment lifecycle, double-booking protection, pricing integrity, and the full RLS/storage policy set directly against the live Supabase project — not mocks. All of that came back clean: the live schema, functions, constraints, triggers, RLS policies, and storage policies match the migration source with **zero drift**.

One genuine issue was found, and it was found empirically, not theoretically: reading the real, single production booking's `booking_status_history` showed an admin had been able to move it through an illegal sequence of statuses — including a completed rental reverting to cancelled, and a cancelled booking silently re-becoming confirmed — because the database never validated that a status change was a legal transition. A fix was written, reviewed, and — following your explicit go-ahead — **applied directly to production and live-tested inside a rolled-back transaction**: all 11 transition tests passed, including exact reproductions of the real production bug (`completed → cancelled` , `completed → confirmed`), which are now correctly rejected. Full details in §11–12.

No code was changed this phase outside that one migration file. 150/150 automated tests still pass, the build is clean, and lint shows the exact same pre-existing warning classes as Phase 4.2 — confirming nothing was accidentally broken while verifying and hardening everything else. **Phase 5 is now fully closed.**

2 Scope & methodology

Per your Phase 5 brief, work proceeded audit-first: read the complete repository, all four existing migration files, both Edge Functions (and their shared `_shared/*.ts` libraries), every domain type, every admin API file, the full checkout flow, and all 28 existing test files before writing or changing anything. Nothing already working was rebuilt.

Live verification used your authenticated Supabase Dashboard SQL Editor session (project `b1suohqmeyzrqycqjdr` , labelled **PRODUCTION**) via browser automation, since this sandbox has no network egress to the live project. Every functional test was wrapped in an explicit `begin; ... rollback;` transaction with results accumulated into a temporary table and printed as a single aggregated report — so behaviour could be proven empirically with **zero permanent writes** to production. This was verified after the fact: `select count(*) from customers/bookings/payments where email like 'phase5-qa-%'` returned **0 rows in all three tables**.

Where the brief's Step 10 called for replaying all migrations against a literal fresh database, this sandbox has no running Docker daemon and no Supabase CLI installed (confirmed via `docker ps` and `which supabase`) — that specific replay could not be executed here. See §17 for how this was substituted and disclosed.

3 Repository & documentation audit

Everything from Phases 0–4.2 was classified. Nothing came back **BROKEN** at the code level; one item came back **MISSING** at the database-behaviour level (§11), and two items are documented as deliberate, in-scope-for-later-phase **FUTURE** items (§13).

Area	Status	Evidence
4 migration files (Phase 0–3)	DONE	Live schema/enum/function/constraint/trigger inventory matches source exactly
<code>create-booking</code> / <code>confirm-payment</code> Edge Functions	DONE	Deployed source fetched live via <code>pg_get_functiondef</code> -equivalent inspection; matches repo bundle
Guest checkout flow (search → details → payment → confirmation)	DONE	Full frontend read; server-side re-validation confirmed independent of client-side UX validation
Admin dashboard, Fleet, Pricing, Bookings, Payments, Complaints, Audit Log, Revenue analytics	DONE	All read-only or RLS-governed plain queries; no parallel business-logic path found anywhere
28 existing test files, 150 tests	DONE	Re-run fresh this phase, unchanged, all passing
Booking status transition validation	MISSING	Confirmed live — see §11
Pending-payment booking auto-expiry	FUTURE (documented)	See §13 — needs a business TTL decision + scheduling infra, out of Phase 5 scope
DB-layer independent price re-verification	FUTURE (documented)	See §13 — currently safe via Edge Function + revoked grants, not currently exploitable

4 Booking lifecycle — live verification

Tested directly against production inside a rolled-back transaction, using the real Suzuki Alto VXS vehicle (`825417ba-7514-4b17-8b8a-ed6ad11b9750` , 50.00 AED/day list, 40.00 AED/day client rate) and the two real DXB locations already in the database.

Test	What was proven	Result
Baseline creation	A valid <code>create_booking()</code> call with correct dates/vehicle/prices succeeds and returns a <code>pending_payment</code> booking + matching <code>pending</code> payment row	PASS
Invalid date range	<code>end_date <= start_date</code> is rejected at the database layer with a clear error, independent of the frontend's own UX-only validation	PASS
Negative price	A negative <code>total_price / unit_price</code> is rejected by the database's non-negative check constraint	PASS
Nonexistent vehicle ID	Booking creation against a vehicle ID that doesn't exist fails with a clear, specific error rather than a generic 500	PASS

Same-day turnover	Two bookings meeting exactly at a shared boundary day are currently blocked by the <i>inclusive</i> daterange design (documented informational finding, not a bug — see note below)	INFORMATIONAL
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Same-day turnover note: `daterange(start_date, end_date, '[]')` (inclusive both ends) means a booking ending on day X and a new booking starting on day X are treated as overlapping by `bookings_no_overlap`, even though in a real rental business a car is often returned and re-rented the same day. This is a genuine business-policy question (does Bliss Rent want to allow same-day turnover, and if so, does that require a same-day buffer/cleaning window?) rather than a bug — flagging it for a future phase's business-rule decision rather than changing exclusion-constraint semantics unasked.

5 Payment lifecycle — live verification

Test	What was proven	Result
Decline idempotency	Once a payment is marked <code>failed</code> , <code>confirm_payment()</code> cannot later flip it to <code>paid</code> — it returns the already-resolved state instead of double-processing	PASS
Paid idempotency	Once a payment is marked <code>paid</code> (and the booking auto-confirmed), <code>confirm_payment()</code> cannot later flip it to <code>failed</code>	PASS
Successful confirmation	A pending payment marked <code>paid</code> correctly auto-transitions its booking from <code>pending_payment</code> to <code>confirmed</code> in the same call	PASS
Nonexistent payment ID	<code>confirm_payment()</code> against a payment ID that doesn't exist fails with a clear, specific error	PASS

Grants were independently confirmed: `create_booking` and `confirm_payment` are `SECURITY DEFINER` functions with `EXECUTE` revoked from `anon` / `authenticated` and granted only to `postgres` / `service_role` — so these guarantees hold specifically because only the Edge Functions (running with the service-role key) can call them, exactly as designed in Phase 2.

6 Availability & double-booking protection — live verification

Test	What was proven	Result
Overlapping booking rejected	Attempting to create a second booking for the same vehicle over dates overlapping an existing non-cancelled booking is rejected by the <code>bookings_no_overlap</code> GiST exclusion constraint	PASS
Non-overlapping booking succeeds	A second, genuinely non-overlapping booking for the same vehicle succeeds normally	PASS
<code>available_vehicles()</code> excludes a reserved vehicle	Searching the exact date range of an existing confirmed booking correctly omits that vehicle from results	PASS
<code>available_vehicles()</code> includes a freed vehicle	Searching a date range with no overlapping bookings correctly includes the vehicle	PASS

This is the most safety-critical guarantee in the system — a real double-booking would mean two customers physically expecting the same car — and it held on every test, enforced at the database layer (not just application code), which is exactly where it needs to live.

7 Pricing integrity / anti-tampering — live verification

The full deployed source of `create_booking()` was fetched directly from the live database (not read from the migration file, to rule out drift) via a `pg_get_functiondef`-style introspection query. It confirmed the function accepts `p_unit_price / p_total_price` as raw parameters from its caller, with only a non-negative check — it does **not** independently recompute the price from the `pricing` table itself.

Tested live: a request with a negative price is rejected (§4); this in itself doesn't test tamper-resistance against a plausible-looking but wrong price, since the database layer trusts its caller for the price value. What actually protects pricing integrity is the layer *above* the database: the `create-booking` Edge Function independently looks up the vehicle's real pricing-ladder rate and computes the total server-side before ever calling `create_booking()` — a browser cannot reach `create_booking()` directly, because its `EXECUTE` grant is revoked from `anon / authenticated` (confirmed live, §5). So the practical guarantee — a customer cannot pay less than the real price — holds today, but it holds because of the Edge Function plus the grants, not because of an independent database-layer check. This is documented as a deliberate defense-in-depth observation in §13, not a currently-exploitable gap.

8 Security & RLS audit — live verification

Full `pg_policies` inventories were pulled live for both the `public` schema (all 13 tables) and the `storage` schema (all 6 bucket policies), and compared line-by-line against the Phase 0 migration source.

Area	Result
All 13 <code>public</code> schema RLS policies	Match migration source exactly — zero drift
Storage bucket public/private flags + all 6 storage policies	Match migration source exactly — zero drift
<code>is_admin()</code> gating on every admin-only table/policy	Confirmed present and correctly referenced
<code>create_booking / confirm_payment</code> grants	<code>EXECUTE</code> revoked from <code>anon / authenticated</code> ; granted only to <code>postgres / service_role</code>
Driver-document storage policy	Admin read-only via signed URL, confirmed — never a public URL

9 Edge Function audit

`create-booking` and `confirm-payment`'s full deployed SQL/TypeScript source was read directly from the live project (not assumed from the repo). Both match the repo's bundled source with zero drift. Both independently re-validate input server-side (date ranges, vehicle existence, pricing lookup) rather than trusting client-supplied values, both return typed `BookingErrorCode`s the frontend maps to clean user-facing messages, and both use wildcard CORS consistent with the rest of the public API surface. The shared `TEST ONLY` simulated payment provider is clearly labelled as such in its own source and is not reachable in a way that could be mistaken for a real payment gateway.

10 Full customer/admin lifecycle consistency check

Rather than performing a fresh synthetic browser click-through this session, the one real production booking already created and confirmed by you in the prior session (documented in `checkout-and-admin-fixes-2026-08-26.md`, reference **BLS-D300AC89**) was read directly and cross-checked across every related table: `bookings`, `customers`, `drivers`, `vehicles`, the

`vehicle_operational_status` view, `payments`, and `booking_status_history`. This is real, non-synthetic evidence of the full stack working together, and it's how §11's finding was actually discovered.

Consistent and correct: the booking's `total_price` (200.00 AED) matches 4 nights × the vehicle's 50.00 AED/day list rate; the payment row shows `status: paid` for the exact same 200.00 AED via the test provider; the vehicle's `vehicle_operational_status` view correctly reports `reserved` (not yet `rented`), which is correct because today's date (Aug 26) is still before the booking's Aug 27 start — `rented` only applies once a booking is `active`. The driver record, customer record, and pickup/dropoff locations all resolve correctly and match what the admin Booking Detail page would render.

Not consistent: the booking's status-history trail — see §11.

11 Finding: booking status transition gap (confirmed live)

What was found. `booking_status_history` for the real booking above contains this actual, real sequence of transitions (each one legitimately logged by the existing audit trigger, timestamps and admin actor ID included):

```
pending_payment → confirmed      (23:33:02 – correct: payment succeeded)
confirmed       → cancelled      (23:49:51 – admin action)
cancelled       → completed      (23:49:57 – ILLEGAL: cancelled is meant to be terminal)
completed      → cancelled      (23:50:07 – ILLEGAL: completed is meant to be terminal)
cancelled      → confirmed      (23:54:12 – ILLEGAL: re-confirms an allegedly-cancelled
booking)
confirmed      → active         (23:54:15 – legal in isolation)
active         → cancelled      (23:55:39 – legal in isolation)
cancelled      → completed      (23:57:36 – ILLEGAL, again)
completed      → confirmed      (23:57:39 – ILLEGAL, again)
confirmed      → completed      (00:36:51 – skips 'active' entirely)
completed      → confirmed      (00:37:21 – ILLEGAL, again)
```

Why this matters. `completed` and `cancelled` are meant to be terminal states in a car-rental lifecycle. This trail proves the database currently allows a completed rental to un-complete itself, and a cancelled booking to silently become confirmed again — while its date range still holds the `bookings_no_overlap` exclusion lock, meaning a "cancelled" booking that never actually frees its dates for another customer is a real, observed possibility, not a hypothetical.

Root cause. `adminBookingsApi.ts`'s `updateBookingStatus()` is a plain RLS-governed `UPDATE` with no state-machine validation — by design, per the Phase 3 comment on that function ("*A plain RLS-governed UPDATE — no Edge Function needed*"), the assumption at the time was that the existing audit trigger firing on every change was sufficient. It logs faithfully; it just never validated legality. This is squarely within this phase's chartered scope — Step 2 of your brief asked to verify the booking lifecycle, and "Business Logic Hardening" is literally in this phase's title.

12 Fix applied and live-tested: booking status state-machine guard

The migration `supabase/migrations/20260828000000_phase5_booking_status_guard.sql` extends the existing Phase 0/3 trigger function `handle_booking_status_change()` **in place** (same trigger binding, new body — the same technique Phase 3 already used to add the `audit_logs` insert) so that it rejects any transition outside the legal set below, before the row is ever written — meaning an illegal transition never reaches `booking_status_history` or `audit_logs` at all. **This has now been applied directly to the production database** and independently confirmed by fetching the live deployed function body back out with `pg_get_functiondef` — it matches the intended source exactly, zero drift.

From	Legal next states
<code>pending_payment</code>	<code>confirmed</code> , <code>cancelled</code>
<code>confirmed</code>	<code>active</code> , <code>cancelled</code>
<code>active</code>	<code>completed</code> , <code>cancelled</code>
<code>completed</code> , <code>cancelled</code>	<i>none — terminal</i>

```
create or replace function handle_booking_status_change()
returns trigger language plpgsql as $$
begin
  new.updated_at := now();
  if tg_op = 'UPDATE' and old.status is distinct from new.status then
    if not (
      (old.status = 'pending_payment' and new.status in ('confirmed', 'cancelled')) or
      (old.status = 'confirmed'       and new.status in ('active', 'cancelled')) or
      (old.status = 'active'          and new.status in ('completed', 'cancelled'))
    ) then
      raise exception 'Illegal booking status transition: % -> %.', old.status, new.status
        using errcode = '23514';
    end if;
    insert into booking_status_history (...) values (...); -- unchanged
    insert into audit_logs (...) values (...);             -- unchanged
  end if;
  return new;
```

```
end;  
$;
```

Blast-radius check performed before writing this: a repo-wide search confirmed only two code paths ever change `bookings.status` — `confirm_payment()` (only ever `pending_payment` → `confirmed`, which is legal under this guard) and the admin's `updateBookingStatus()` (now guarded).

No frontend change was required — `BookingDetailPage.tsx` already wraps the update call in try/catch and renders the database's error message in its existing `updateError` banner, with no optimistic UI update to roll back, so a rejected transition simply shows a clear message and leaves the booking's displayed status unchanged.

Live test results — 11/11 PASS

Run directly against production inside a single `begin; ... rollback;` transaction, using two disposable test bookings (customer `phase5-qa-guard@example.com`) against the same real Suzuki Alto VXS vehicle, on date ranges far from the real booking (no overlap risk). Nothing from this test persisted — confirmed after the fact (see below).

#	Test	Expected	Actual	Result
T13	<code>pending_payment</code> → <code>active</code>	blocked	Illegal booking status transition: <code>pending_payment</code> → <code>active</code> .	PASS
T14	<code>pending_payment</code> → <code>confirmed</code>	allowed	success	PASS
T15	<code>confirmed</code> → <code>pending_payment</code>	blocked	Illegal booking status transition: <code>confirmed</code> → <code>pending_payment</code> .	PASS
T16	<code>confirmed</code> → <code>active</code>	allowed	success	PASS
T17	<code>active</code> → <code>completed</code>	allowed	success	PASS
T18	<code>completed</code> → <code>cancelled</code> (exact repro of the real prod bug)	blocked	Illegal booking status transition: <code>completed</code> → <code>cancelled</code> .	PASS
T19	<code>completed</code> → <code>confirmed</code> (exact repro of the real prod bug)	blocked	Illegal booking status transition: <code>completed</code> → <code>confirmed</code> .	PASS
T20	<code>completed</code> → <code>completed</code> (same-status no-op)	allowed	success	PASS
T21	<code>completed</code> → <code>pending_payment</code>	blocked	Illegal booking status transition: <code>completed</code> → <code>pending_payment</code> .	PASS
T22	<code>confirmed</code> → <code>cancelled</code> (second test booking)	allowed	success	PASS
T23	<code>cancelled</code> → <code>active</code> (terminal-state check)	blocked	Illegal booking status transition: <code>cancelled</code> → <code>active</code> .	PASS

Post-test verification: `select count(*) from customers where email like 'phase5-qa-%'` → 0; the two disposable test booking IDs → 0 rows (both gone with the rollback). The real production booking (`d300ac89-...`, ref BLS-D300AC89) was independently re-checked and is untouched: still

`status = confirmed`, still exactly **11** rows in its `booking_status_history` — identical to before this migration was applied.

Deployment sequence used, per your explicit instruction: (1) the `create or replace function` statement was run standalone against production; (2) the 11-test live-verification script above was run and rolled back; (3) residue and the real booking were re-checked; (4) the deployed function body was pulled back with `pg_get_functiondef` and diffed against the intended source — exact match. All four steps completed with no errors and no anomalies.

13 Other findings — documented, deliberately postponed

Per your explicit scope rule ("if you discover something belonging to a later phase, document it and leave it for that phase, unless it is required to make the current backend flow function correctly"), these two items are recorded here rather than fixed now:

Finding	Why it's not fixed now
Pending-payment bookings never expire — a booking that's created but never paid for stays <code>pending_payment</code> forever, holding its date range against the <code>bookings_no_overlap</code> constraint indefinitely.	Fixing this needs a business decision (how long is the grace window?) and scheduling infrastructure (e.g. <code>pg_cron</code>) — the brief explicitly lists "advanced deployment infrastructure" as out of scope for this phase.
The database layer trusts the price it's given by <code>create_booking()</code> 's caller rather than independently recomputing it from the <code>pricing</code> table.	Not currently exploitable — the Edge Function computes the real price server-side, and the function's <code>EXECUTE</code> grant is revoked from every role a browser could authenticate as. Adding a second, independent price check inside the database would be genuine defense-in-depth, but the brief's language ("if a <i>real</i> gap exists, fix it") points at gaps that are actually reachable today, and this one isn't.

14 Automated test suite

No new test files were added this phase. The one genuine gap found (§11) lives entirely at the database/trigger layer — it has no equivalent pure-function logic in the frontend to unit-test, and the existing suite already has a generic test proving any Postgres error (including a raised exception) correctly surfaces as an `AdminApiError` with the database's message (`adminBookingsApi.test.ts`). The actual proof this fix works was done at the live-database level instead (§12's 11-test table) — a stronger guarantee than a mocked unit test could give, since it exercises the real deployed trigger.

150 / 150 PASSING — re-run fresh **after** the migration was applied to production, 28 test files, ~40s, unchanged from Phase 4.2 (expected: this fix lives entirely in the database, not in any file these tests touch).

15 Build & lint results

`npm run build ( tsc -b && vite build )`: **PASS** — zero TypeScript errors, only the pre-existing "chunk larger than 500 kB" advisory (unrelated to correctness, present since Phase 4).

`npm run lint ( oxlint )`: **PASS** — the same warning classes as Phase 4.2 (`set-state-in-effect` , one `exhaustive-deps` , one `only-export-components`), zero new warnings — confirming no source files were modified this phase beyond the one migration file.

`git status` : **CLEAN** aside from the new migration file and this report — no unintended edits anywhere in the tree.

16 Migration & schema drift verification

Live introspection of tables, views, enums, functions, constraints, triggers, RLS policies, storage policies, and grants was compared field-by-field against all four existing migration files.

Zero drift found anywhere. This directly addresses the root-cause pattern from the prior bug-fix session (documented in `checkout-and-admin-fixes-2026-08-26.md`): every earlier outage traced back to a migration file that existed in the repo but had never actually been run against the live database. That is no longer the case for Phases 0–3 as of this audit.

17 Environment limitation disclosed

Step 10 of the brief (fresh-database migration replay) could not be executed in this sandbox.

Docker is installed but its daemon is not running (`docker ps` fails with "no such file or directory"), and the Supabase CLI is not installed. A literal "spin up an empty Postgres, run all 5 migrations in order, confirm success" replay is therefore not possible here.

Substitute evidence provided instead: the live schema/RLS/grant/function-body drift check in §16, which arguably verifies the more important property — that what's actually deployed matches what the repo says should be deployed — directly addressing the exact failure mode (migrations existing but never being run) that caused every real outage in this project so far.

If you want the literal fresh-DB replay done, it needs an environment with a working Docker daemon and the Supabase CLI (`supabase start` , then `supabase db push` against the local instance) — either on your own machine, or a future session where those are available.

18 Files changed this phase

File	Change
<code>supabase/migrations/20260828000000_phase5_booking_status_guard.sql</code>	New — booking status state-machine guard (§12). Applied to production and live-tested (11/11 PASS).

Nothing else in the repository was modified this phase — no frontend files, no other backend files, no test files. This is by design: the audit found the codebase to already match its own documentation and tests, so nothing needed rebuilding except this one confirmed gap.

19 Production-readiness checklist

Area	Result
Booking creation, validation, error handling	PASS
Payment lifecycle & idempotency	PASS
Double-booking / availability protection	PASS
Pricing integrity (Edge Function + grants)	PASS
RLS & storage security	PASS
Edge Function correctness & drift	PASS
Live schema/migration drift	PASS
Fresh-DB migration replay (Step 10)	NOT EXECUTED — sandbox limitation, see §17
Automated tests / build / lint / git status	PASS (150/150, clean, unchanged warnings, clean tree)
Booking status transition integrity	PASS — guard deployed & live-tested, 11/11 (§11–12)
Pending-payment expiry policy	Not implemented — documented, deferred (§13)

20 Phase 5 status: CLOSED

✓ Every Phase 5 checklist item is PASS except the one disclosed, substituted sandbox limitation (§17) and the two deliberately-postponed future items (§13) — both explicitly out of this phase's scope per your own brief. **Phase 5 is complete.**

Per your instruction: Steps 1–4 of the closing workflow are done — migration applied, live-tested (11/11), full baseline re-confirmed (150/150 tests, clean build, unchanged lint, clean git tree), zero production test-data residue. This report is Step 5's deliverable.

Per the project's standing instructions: stopping here and awaiting your explicit confirmation before any Phase 6 work begins — no WhatsApp integration, deployment infrastructure, or new features have been started or will be started until you give the go-ahead. Anything already DONE in Phases 0–5 stays DONE; Phase 6 will only build what its own audit finds actually missing.